

BETA EXPANSION

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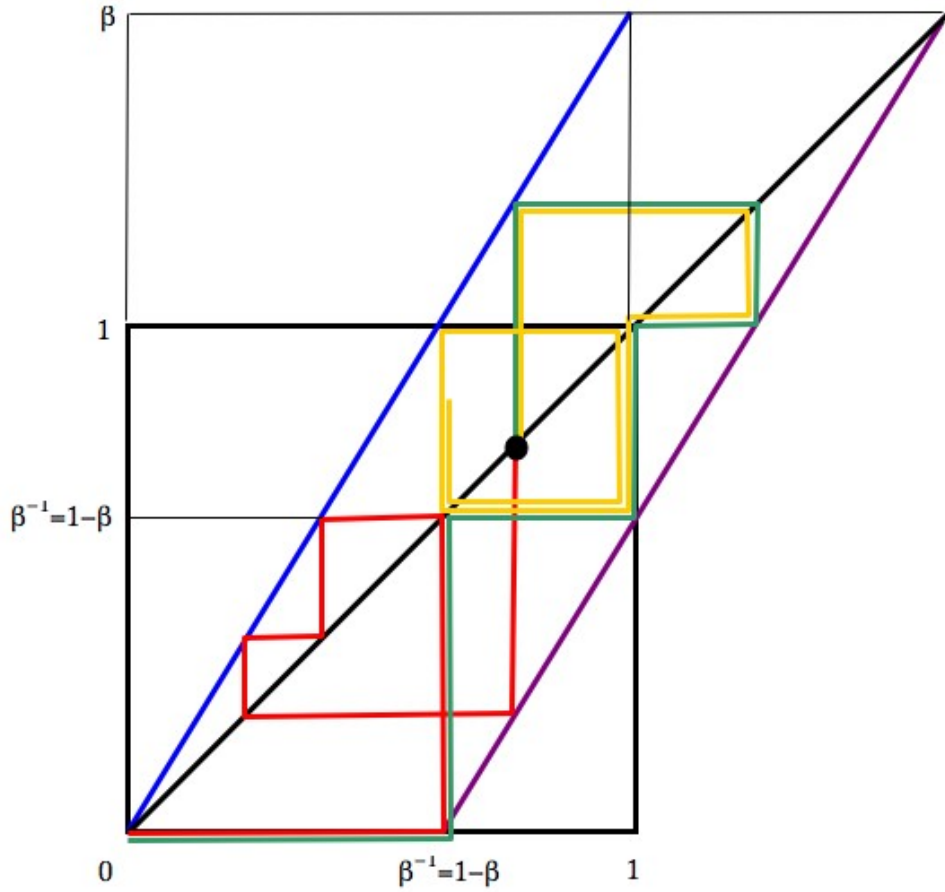


FIGURE 1. Beta expansions of the golden mean.

In the diagram, $\beta = (1 + \sqrt{5})/2 \approx 1.618$ satisfies $\beta^2 - \beta = 1 = 0$ and is the golden mean. The black dot is the point $x = 3 - \sqrt{5} \approx .764$. Three β -expansions of x are shown.

- (1) The red expansion is the *greedy* expansion and gives

$$x = \beta^{-1} + \beta^{-4} = 10010^\omega.$$

- (2) The green expansion gives

$$x = \beta^{-2} + \beta^{-3} + \beta^{-4} = 01110^\omega.$$

- (3) The yellow expansion is the *lazy* expansion and gives

$$x = \beta^{-2} + \beta^{-3} + \beta^{-4} = 100(01)^\omega.$$

Remarks:

- (1) From any $x \in [0, 1)$ on the diagonal, one moves vertically to either the blue or violet line corresponding to the lines of slope β through 0 and β on the x -axis.

Then move horizontally to the diagonal. This gives $T(x)$.

- (2) For the greedy expansion one always stays in the box $[0, 1) \times [0, 1)$ and there is a unique choice of vertical move. Equivalently, for $x \in [0, 1)$ one chooses

$$a(x) = [\beta x], \quad T(x) = (\beta x) = \beta x - [\beta x].$$

- (3) For other allowable expansions note the diagonal and the crimson line cross at $x = \beta$, and in general when

$$x = \beta \left(x - [\beta] \beta^{-1} \right), \quad \text{i.e. } x = \frac{[\beta]}{\beta - 1}.$$

From geometric considerations the allowable values of $a(x)$ are those integers such that

$$0 \leq \beta - a(x) \leq \frac{[\beta]}{\beta - 1}.$$

(One needs to slide the blue line $x \mapsto \beta x$ down by an integer amount so that it is below the level $\frac{[\beta]}{\beta - 1}$.) That is

$$\begin{aligned} \beta x - \frac{[\beta]}{\beta - 1} &< a(x) \leq \beta x, \\ \left\lceil \beta x - \frac{[\beta]}{\beta - 1} \right\rceil + 1 &\leq a(x) \leq [\beta x]. \end{aligned}$$

(C.f. [Barat et al; p2003 (2)]. Note that if β is not an integer then $[\beta] = \lceil \beta \rceil$ and note that “ $\lceil$ ” and “ $\rceil$ ” are used in the above reference.)

The value of $T(x)$ is then

$$T(x) = \beta x - a(x),$$

which may be greater than 1.

- (4) For a proof of the Parry characterisation of the $d\beta$ -shift see [Lot02, Theorem 7.2.9, p211] (under Frougny, Numerations Systems, in my “Papers” collection).

REFERENCES

- [Lot02] M. Lothaire, *Algebraic combinatorics on words*, Encyclopedia of Mathematics and its Applications, vol. 90, Cambridge University Press, 2002. A collective work by Jean Berstel, Dominique Perrin, Patrice Seebold, Julien Cassaigne, Aldo De Luca, Steffano Varricchio, Alain Lascoux, Bernard Leclerc, Jean-Yves Thibon, Veronique Bruyere, Christiane Frougny, Filippo Mignosi, Antonio Restivo, Christophe Reutenauer, Dominique Foata, Guo-Niu Han, Jacques Desarmenien, Volker Diekert, Tero Harju, Juhani Karhumaki and Wojciech Plandowski; With a preface by Berstel and Perrin.